How many street-view images do we actually need?

Phase-2 attention pruning — Qwen3.6-27B, $k = 1, 2, 3$, $n = 1663$ solved questions

The goal

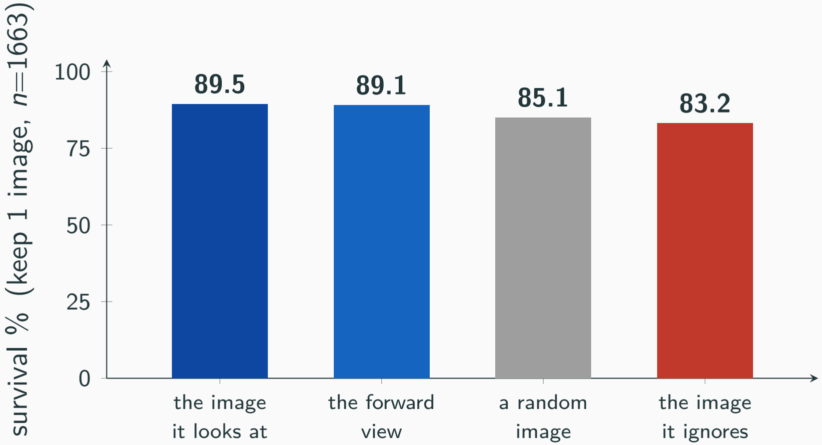
Every question feeds the model **4 street-view images** of the same spot: the **forward view**, the back view, and the two side views.

What we want to know
Can we **throw most of them away**? If one image already gives the right answer, feeding four is wasted compute.

How we test it. Solve each question on all 4 images, see which image the model *looks at* most, then keep only the top k images and ask again.

Survival % $=$ of the questions it got right on 4 images, how many it *still* gets right on the smaller set.

One image is (almost) enough



Keep just the **one image the model looks at**, drop the other three, and the answer stays right **9 times out of 10**.

Does the *choice* of image matter?

- A random single image already survives **85%**. So a lot of the answer isn't really about *which* image you keep.
- But the image the model **looks at** does a bit better: **89.5%** vs **85%** — a real +4.4 **pt** edge over a random pick.
- And the image it **ignores** is worst (83%) — so the model's attention is pointing at something useful, not noise.

On the harder questions — the ones that genuinely need to look at an image — the edge of the attended image over a random one grows to +8.8 **pt**.

The full numbers — every arm, every k

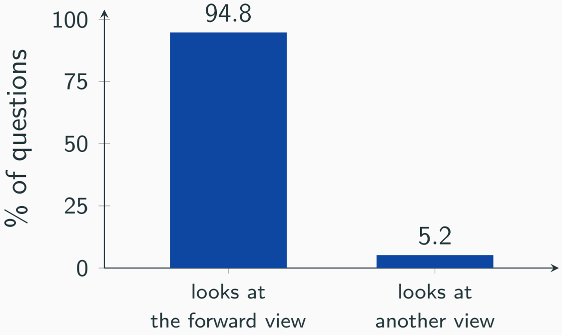
keep →	**1 image**	**2 images**	**3 images**
images it looks at	**89.5**	**92.7**	**94.8**
random images (averaged)	85.1	90.9	93.0
images it ignores	83.2	88.7	91.2
edge (looks-at − random)	+4.4	+1.8	+1.8

The 3 random draws agree — "random" is averaged over 3 seeds, each landing in the same place (85% isn't a lucky draw):

random seed	keep 1	keep 2	keep 3
3407	85.9	90.3	93.6
101	85.1	91.1	92.7
202	84.4	91.2	92.9

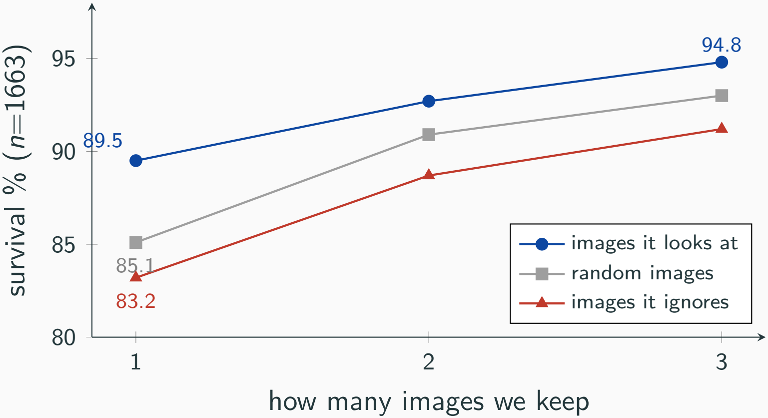
survival % ; n=1663 per arm (random n=1663/seed, 4989 pooled).

And the image it looks at is the forward view



- Almost always (**94.8%**) the image the model looks at is the **forward view** — the one pointing down the street.
- So "keep the one it looks at" and "just always keep the forward view" give the same answer: **89.5%** vs **89.1%**.
- **Simple rule:** keep the forward view, drop the other three.

If you keep more images, it helps — a little



Keeping the **looked-at** images always wins, but as you keep more, it stops mattering *which* ones — by 3 images everything is close. The real saving is at **1 image**.

The forward view wins in every topic

topic	forward view	random image	edge
road surface	97.5	92.5	+4.9
junction type	91.2	91.1	+0.1
transit density	90.8	86.2	+4.6
road type	89.4	85.6	+3.8
urban density	86.9	81.4	+5.5
land use	87.1	82.3	+4.9
building height	85.8	79.9	+6.0
amenity richness	77.2	74.7	+2.5
all	**89.1**	**85.1**	**+3.9**

Keeping just the **forward view** beats keeping a random image in **every topic** — the answer lives in the forward view everywhere, not just on average.

Take-home

- ✓ We **don't need all 4 street-view images** — **one** is enough (89.5% of answers survive).
- ✓ That one image is the **forward view**: it's what the model looks at 94.8% of the time, so just always keep it.
- ✓ The looked-at image beats a random one by +4.4 **pt** overall (+8.8 **pt** on the questions that really need an image) — attention is pointing at something real, in every topic.

3–4× fewer image tokens per question, same answer.

Repro: `python attention_prune/analyze_prune.py` | k=1, 2, 3, 3-seed random, n=1663.